

MILLBROOK 3 W, NY, USA

WMO: 743690

Lat: 41.7857N

Lon: 73.7422W

Elev: 126

StdP: 99.82

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 64756

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.4	-14.3	-23.6	0.5	-16.1	-20.7	0.6	-13.4	4.0	2.5	3.6	0.2	1.0	N/A	0.411

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	31.3	22.8	29.9	21.9	28.5	21.2	24.2	29.2	23.4	27.8	22.5	26.6	1.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	17.6	27.1	21.9	16.8	26.0	21.2	16.1	25.0	73.5	29.3	70.0	27.9	66.9	26.7	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 3.3	3.0	2.6	DB	-22.2	34.2	3.3	1.3	-24.6	35.2	-26.5	36.0	-28.4	36.7	-30.8	37.7	(4)
(5)			WB	-23.3	25.9	3.4	1.1	-25.8	26.7	-27.8	27.3	-29.7	27.9	-32.2	28.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	-3.2	-2.3	2.2	9.0	14.7	18.9	21.9	20.7	17.1	11.0	4.8	-0.2
	DBStd	9.93	6.14	5.73	5.51	4.67	4.43	3.42	2.81	2.93	4.02	4.58	4.76	5.29
	HDD10.0	1613	411	344	251	72	8	0	0	0	1	43	166	318
	HDD18.3	3489	668	577	502	282	132	33	4	10	71	231	405	575
	CDD10.0	1478	1	1	8	43	153	268	371	333	214	75	11	2
	CDD18.3	312	0	0	0	3	19	51	117	84	34	4	0	0
	CDH23.3	2763	0	0	2	48	218	463	1029	693	284	24	0	0
	CDH26.7	768	0	0	0	13	54	119	324	182	72	3	0	0
Wind	WSAvg	1.1	1.3	1.4	1.4	1.5	1.2	0.9	0.9	0.8	0.9	1.1	1.2	1.2
Precipitation	PrecAvg	1216	86	73	103	95	93	108	114	121	114	121	94	105
	PrecMax	1601	188	215	169	199	170	213	261	266	284	443	175	192
	PrecMin	833	35	32	22	29	31	18	35	61	22	21	28	28
	PrecStd	171	40	37	35	41	37	57	54	53	72	91	36	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.3	16.7	22.1	29.0	30.5	32.2	33.7	32.3	31.4	26.0	20.4	16.8
		MCWB	13.5	12.2	14.0	16.2	20.8	23.6	24.3	25.2	23.2	19.4	15.0	14.9
	2%	DB	11.6	12.6	17.5	23.9	28.2	29.8	31.6	30.3	28.9	23.2	17.6	13.0
		MCWB	9.8	8.5	11.5	15.5	19.9	21.8	23.0	22.8	21.3	18.1	14.1	10.7
	5%	DB	7.7	9.0	13.6	21.0	25.9	27.8	30.0	28.7	26.6	21.1	15.5	10.2
		MCWB	5.4	5.3	8.4	13.3	18.3	20.2	22.3	21.6	20.3	17.0	11.4	8.0
	10%	DB	5.3	5.7	10.8	18.2	23.3	26.1	28.4	27.2	24.6	19.2	13.2	7.6
		MCWB	2.9	2.7	6.7	11.9	16.9	19.2	21.7	20.8	19.4	15.5	9.6	5.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.5	14.1	15.7	18.6	22.9	24.5	25.5	25.2	24.2	21.1	16.7	14.9
		MCDB	14.6	17.3	22.0	22.3	28.3	30.2	31.9	30.2	29.1	23.8	19.5	16.3
	2%	WB	9.2	10.8	13.1	16.7	21.4	22.7	24.4	23.9	22.9	19.6	14.7	11.6
		MCDB	10.3	13.0	17.9	21.3	26.7	27.3	29.8	28.0	26.8	21.7	17.0	12.7
	5%	WB	5.7	6.3	10.0	14.7	20.0	21.6	23.6	22.7	21.7	18.0	12.2	9.0
		MCDB	7.4	10.4	13.6	18.5	24.6	25.6	28.3	26.3	25.1	20.9	14.7	10.6
	10%	WB	3.3	4.0	7.2	12.3	18.7	20.4	22.8	21.8	20.6	16.1	9.7	6.6
		MCDB	5.0	6.6	10.6	16.5	22.4	24.1	27.0	25.3	23.6	18.6	12.7	8.1
Mean Daily Temperature Range	5% DB	MDBR	9.2	10.0	10.7	12.7	12.4	11.9	11.9	11.8	12.2	11.3	10.4	8.5
		MCDBR	11.3	13.3	15.3	17.8	15.3	14.0	13.5	13.0	13.8	13.5	13.2	10.6
		MCWBR	10.0	9.8	9.1	9.0	7.8	7.7	6.1	6.2	7.3	7.9	9.7	9.0
	5% WB	MCDBR	11.2	13.0	13.7	12.6	12.5	12.2	11.2	10.6	11.3	10.6	11.9	10.0
		MCWBR	10.3	10.1	9.2	8.3	6.9	7.1	5.7	5.7	6.4	7.5	9.9	9.5
Clear-Sky Solar Irradiance	taub	0.300	0.318	0.341	0.364	0.405	0.461	0.474	0.452	0.378	0.350	0.319	0.299	
	taud	2.400	2.366	2.395	2.383	2.295	2.173	2.147	2.200	2.384	2.472	2.513	2.493	
	Ebn at Noon	847	886	906	906	871	821	804	810	855	841	825	823	
	Edh at Noon	80	97	106	115	129	147	149	137	106	87	72	68	
All-Sky Solar Radiation	RadAvg	1.75	2.62	3.64	4.63	5.21	5.66	5.87	5.17	4.25	2.80	1.94	1.42	
	RadStd	0.13	0.25	0.29	0.50	0.52	0.52	0.35	0.31	0.40	0.29	0.18	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+105	+55

Nomenclature: See separate page